What is kubelet?

Professional Answer

kubelet is an agent running on each worker node that ensures containers are running as defined in Pod specifications.

Unprofessional Answer

kubelet runs pods.

What is kubelet and what are its responsibilities?

kubelet is an agent that runs on every worker node in the cluster.
It receives pod specifications from the API server.
kubelet ensures containers are created and running as defined in the pod.
It monitors container health using probes and restarts containers if needed.
kubelet also reports node and pod status back to the control plane.

What kubelet is internally

- Long-running daemon
- Runs on every worker node

Internal Responsibilities

1. **Pod Lifecycle Management**
    - Reads PodSpec
    - Creates containers
    - Starts/stops containers
2. **Container Health**
    - Executes liveness/readiness/startup probes
3. **Node Status Reporting**

3. **Node Status Reporting**

   - Sends heartbeats to API server
   - Reports resource usage

4. **Volume Management**
   - Mounts/unmounts volumes

kubelet does NOT

Does not Schedule pods
Does not Expose services

What is kubelet?

Professional Answer

kubelet is the primary node agent that runs on every worker node.
It ensures that containers described in PodSpecs are running and healthy.
kubelet communicates with the API Server to receive Pod definitions.
It interacts with the container runtime using CRI.
kubelet reports node and Pod status back to the control plane.

Weak Answer

kubelet runs pods.

Where does kubelet run and how?

Professional Answer

kubelet runs as a system service on every worker node.
It is not deployed as a Pod.
It starts automatically when the node boots.
kubelet manages Pods locally on that node.
If kubelet stops, the node becomes NotReady.

How does kubelet communicate with Kubernetes?

Professional Answer

kubelet communicates with the API Server over HTTPS.
It authenticates using certificates or tokens.
It watches assigned Pods and node configuration.
It sends heartbeats and Pod status updates.
This communication is continuous and bidirectional.

Does kubelet run containers directly?

Professional Answer

No, kubelet does not run containers directly.
It uses the Container Runtime Interface (CRI).
The runtime can be containerd or CRI-O.
kubelet sends instructions to the runtime.
The runtime handles image pulls and container execution.

What responsibilities does kubelet have?

Professional Answer

kubelet ensures Pods are running as defined.
It performs container health checks.
It mounts volumes into Pods.
It enforces CPU and memory limits.
It reports node health and resource usage.